

CADENCE: Real-Time Sustained Stable Item Lookup in High-Speed Data Streams

Supplementary Material

This document is the supplementary material for CADENCE. It collects the full proofs, additional analytical results, and the measurements deferred from the main paper, and is self-contained for the theory: Sections I and II recall the SSI definition, the assumptions, and the main results, and the appendix sections give the deferred material. References to algorithms, sections and figures of the main paper carry the numbering used there; an unqualified table number is this document’s own.

I. PRELIMINARIES

We briefly recall the problem setup; the data structure and the `Update/Query` algorithms are exactly as in the main paper. Table I summarizes the notation of the main paper and of the analysis continued here.

TABLE I
SUMMARY OF NOTATION.

Symbol	Meaning
E	epoch length (updates per epoch)
P	continuity requirement (consecutive epochs)
$\mu_{\min}, \tau$	significance (mean-rate) and stability (MAD) thresholds
$x_t(u)$	per-epoch activity of item u in epoch t (arrival count; weight in the variant)
$\mu_t(u), \text{MAD}_t(u)$	exact P -epoch mean and MAD (Eqs. (2), (3))
m, d	number of rows; buckets per row
$B(i, j), h_i$	bucket at row i , column j ; row- i hash
K, T, C	per-bucket item key, epoch tag, accumulator
S, μ, mad	per-bucket streak; EWMA mean and MAD estimators
α	EWMA smoothing factor
λ	eviction-sharpness parameter ($p_{\text{evict}} = 2^{-\lambda \text{ score}}$)
$\text{score}, p_{\text{evict}}$	replacement score and eviction probability
Q, t_s	qualification streak (consecutive SSI verdicts); run start tag
D, L	DSSI durability (run length); durability threshold
$D_t(u), \Delta_t(u)$	second-order dispersion and centering drift of the window (Theorem 3)
$\kappa_P(\alpha), \theta$	kernel-mismatch constant (5); $\theta = (1 - \alpha)^P$
$B_{\max}, \Xi_P(\alpha)$	global maximum per-epoch quantity and the kernel gap it multiplies in the scale-dependent bound
$S_t^*(u)$	exact continuity streak of u at epoch t (Theorem 1)
Δ_μ, J	block drift and horizon of Corollary 1
C_u, q_0	candidate selections of item u and their score floor (Theorem 5)
A_u, W	updates issued by u over an interval of W epochs (Theorem 6)
ε	admission-override probability (Theorem 6); CADENCE uses $\varepsilon = 0$
$\varepsilon_\mu, \varepsilon_{\text{mad}}$	estimation margins on the mean and MAD gates (Corollary 3)

The stream is partitioned into consecutive *epochs* of E updates. For item u , its per-epoch activity $x_t(u)$ is the number of updates of u in epoch t (or, in the weighted variant, the total weight):

$$x_t(u) \triangleq |\{r \in \text{epoch } t : u(r) = u\}|. \quad (1)$$

Definition 1 (Sustained Stable Item). Given P (number of consecutive epochs), $\mu_{\min}$ (significance threshold), and τ (stability threshold), an item u is a *Sustained Stable Item* (SSI) at epoch t if the following hold over the most recent P epochs: (1) *continuity*, $x_{t-i}(u) > 0$ for all $i \in \{0, \dots, P-1\}$ (a streak $\geq P$); (2) *significance*,

$$\mu_t(u) \triangleq \frac{1}{P} \sum_{i=0}^{P-1} x_{t-i}(u) \geq \mu_{\min}; \quad (2)$$

and (3) *stability*,

$$\text{MAD}_t(u) \triangleq \frac{1}{P} \sum_{i=0}^{P-1} |x_{t-i}(u) - \mu_t(u)| \leq \tau. \quad (3)$$

II. ANALYSIS: ASSUMPTIONS AND MAIN RESULTS

We analyze complexity, the exactness of the continuity descriptor, the approximation error of the significance and stability estimators, and the robustness of retention under hash collisions, stating each result with the key step of its proof; full proofs, the derivations behind the constants, and secondary results (admission guarantee, asymptotic EWMA bound, occupancy lemma, end-to-end detection guarantee, durability corollary) are in the appendix of this document.

A. Time and Space Complexity

Each `Update` probes one bucket per row and does constant work per probe (a rollover committing at most one epoch, a key comparison, a score), so it costs $O(m) = O(1)$ with $m = 2$ and no loop over items or epochs. A `Query` costs $O(md)$, scanning all buckets once. The table holds md fixed-size buckets and no per-item history, so the footprint is $\Theta(md)$, independent of the stream's cardinality.

B. Preliminaries and Assumptions

For an item u , let $x_t(u)$ be its per-epoch quantity (by default the arrival count, (1)), with exact P -epoch mean $\mu_t(u)$ and mean absolute deviation $\text{MAD}_t(u)$ of (2)–(3) as ground-truth targets. When u is tracked we write $\hat{S}_t(u)$, $\hat{\mu}_t(u)$, $\widehat{\text{mad}}_t(u)$ for the descriptors CADENCE maintains, initialized at insertion to $\hat{\mu}_0(u) = \mu_{\min}$ and $\widehat{\text{mad}}_0(u) = 0$; results are stated at a query after epoch t is committed.

Assumption 1 (Periodic synchronization). *Every tracked bucket is synchronized at least once per epoch, either by a lazy rollover when it is accessed during `Update` (Algorithm 1) or by the periodic `Query` sweep (Algorithm 2), so that each completed epoch is committed exactly once to the descriptors of the item then residing in the bucket.*

Assumption 1 is the only global assumption. Every other hypothesis is local and stated where it is used: continuous tracking in Theorems 1–3, continuous tracking, bounded dispersion and bounded block drift in Corollary 1, the three gates in Theorem 4, and a score floor in Theorem 5. We assume no bound on the per-epoch quantity: the estimator bounds are stated in terms of each item's own dispersion, so the stream's dynamic range never enters them.

C. Exactness of the Continuity Descriptor

Define the exact continuity streak of item u at epoch t as $S_t^*(u) \triangleq \max\{\ell \geq 0 : x_{t-i}(u) > 0, \forall i \in \{0, \dots, \ell - 1\}\}$.

Theorem 1 (Exact streak, conditional on continuous tracking). *Suppose item u is tracked by CADENCE from epoch $t - \ell + 1$ through epoch t for some $\ell \geq S_t^*(u)$, that is, no collision evicts it during that interval. Then, under Assumption 1, the streak counter stored by CADENCE is exact: $\hat{S}_t(u) = S_t^*(u)$.*

Proof idea. Induction on committed epochs: the rollover (Algorithm 1, Line 8) applies exactly the recurrence that defines S_t^* , and Assumption 1 rules out skipped or double commits. The hypothesis is what collisions threaten, since an eviction resets the streak: the retention policy of Section II-F exists to make it hold, and Theorem 4 shows that a qualified item cannot be evicted by a collision while it stays qualified.

D. Approximation Error of the Mean Estimator

For a tracked item u , CADENCE updates $\hat{\mu}_t(u) = (1 - \alpha)\hat{\mu}_{t-1}(u) + \alpha x_t(u)$ with $\alpha \in (0, 1)$; unrolling P steps, valid because each epoch is committed exactly once, gives

$$\hat{\mu}_t(u) = (1 - \alpha)^P \hat{\mu}_{t-P}(u) + \alpha \sum_{k=0}^{P-1} (1 - \alpha)^k x_{t-k}(u) \quad (4)$$

an exponentially weighted average, whereas Definition 1 asks for the uniform average over the last P epochs. Bounding every per-epoch value by the stream maximum $B_{\max}$ bounds the gap by $B_{\max} \Xi_P(\alpha)$, where $\Xi_P(\alpha) \triangleq \sum_{k < P} |\alpha(1 - \alpha)^k - \frac{1}{P}| + (1 - \alpha)^P$. This bound is correct but vacuous at the operating point: $\Xi_{12}(0.1) = 0.61$ with $B_{\max} = 152$ on WebDocs and 5,010 on CAIDA, against $\mu_{\min} = 8$. It is loose because it discards what the task supplies: an item with small dispersion has small per-epoch deviations, and they sum to zero. Writing $c_k \triangleq \alpha(1 - \alpha)^k - \frac{1}{P}$, define

$$\kappa_P(\alpha) \triangleq \frac{1}{2} \left(\max_{k < P} c_k - \min_{k < P} c_k \right) = \frac{\alpha}{2} (1 - (1 - \alpha)^{P-1}) \quad (5)$$

the closed form following because c_k decreases in k ; at $P = 12, \alpha = 0.1$, $P\kappa_P(\alpha) = 0.41$.

Theorem 2 (Mean estimation error). *Under Assumption 1, for any item u tracked throughout epochs $t - P, \dots, t$, so that the value committed for each epoch of the window is the item's full per-epoch quantity,*

$$\begin{aligned} |\hat{\mu}_t(u) - \mu_t(u)| &\leq P \kappa_P(\alpha) \text{MAD}_t(u) \\ &\quad + (1 - \alpha)^P |\hat{\mu}_{t-P}(u) - \mu_t(u)| \end{aligned} \quad (6)$$

TABLE II

MEASURED ERROR AGAINST THE BOUNDS OF THEOREMS 2 AND 3, ON THE EXACT PER-EPOCH SERIES OF THE 2,000 MOST FREQUENT ITEMS PER TRACE. EACH PAIR IS MEAN ERROR / MEAN BOUND; THE LAST COLUMN OF EACH BLOCK IS THE FRACTION OF ITEM-EPOCHS ON WHICH THE BOUND FALLS BELOW ITS OWN GATE, $\mu_{\min}$ OR τ .

Trace	item-epochs	mean, Thm. 2		MAD, Thm. 3	
		error / bound	$< \mu_{\min}$	error / bound	$< \tau$
CAIDA	446,467	0.38 / 1.23	97.1%	0.38 / 2.06	81.9%
Campus	420,160	0.45 / 1.61	98.5%	0.39 / 2.55	62.0%
Zipf	547,958	0.40 / 1.45	98.7%	0.30 / 2.24	75.2%
WebDocs	1,658,032	0.62 / 1.69	99.0%	0.45 / 3.13	52.6%

and, whenever $\mu_t(u) \geq P \text{MAD}_t(u)/2$ so that the extremal series is non-negative, the first term is attained by deviations supported on $\arg \max_k c_k$ and $\arg \min_k c_k$.

Proof idea. Since $\sum_{k < P} c_k = -(1 - \alpha)^P$, substituting $x_{t-k} = \mu_t + \delta_k$ with $\sum_k \delta_k = 0$ into (4) gives the exact identity $\hat{\mu}_t - \mu_t = (1 - \alpha)^P (\hat{\mu}_{t-P} - \mu_t) + \sum_{k < P} c_k \delta_k$. The zero-sum constraint forces the positive and negative parts of δ to carry mass $Z/2$ each, where $Z = \sum_k |\delta_k| = P \text{MAD}_t(u)$, so $|\sum_k c_k \delta_k| \leq \frac{Z}{2} (\max_k c_k - \min_k c_k)$, with equality when the two parts sit on $\arg \max_k c_k$ and $\arg \min_k c_k$; a Hölder bound that ignores the constraint would cost a factor 1.51.

For an item with $\text{MAD}_t(u) \leq \tau = \sqrt{5}$ the first term is at most 0.93. The trailing term is not a transient: $(1 - \alpha)^P = 0.28$ is the kernel weight outside the window and $|\hat{\mu}_{t-P} - \mu_t|$ is data-dependent, so Theorem 2 alone yields no number; iterating the identity over P -epoch blocks does, for a resident item, under two hypotheses on its exact statistics.

Corollary 1 (Stable regime). *Write $\theta = (1 - \alpha)^P$. Suppose that u is tracked throughout epochs $t - JP, \dots, t$; that its true dispersion satisfies $\text{MAD}_{t-jP}(u) \leq \tau$ for $0 \leq j < J$; and that its block means drift by at most Δ_μ , i.e. $|\mu_{t-(j+1)P}(u) - \mu_{t-jP}(u)| \leq \Delta_\mu$. Then*

$$\begin{aligned} |\hat{\mu}_t(u) - \mu_t(u)| &\leq \frac{P \kappa_P(\alpha) \tau + \theta \Delta_\mu}{1 - \theta} \\ &\quad + \theta^J |\hat{\mu}_{t-JP}(u) - \mu_{t-JP}(u)|. \end{aligned}$$

Proof idea. Apply the identity at epoch $t - jP$ and insert μ_{t-jP} into its trailing term, giving $e_j \leq P \kappa_P(\alpha) \tau + \theta(e_{j+1} + \Delta_\mu)$ for $e_j \triangleq |\hat{\mu}_{t-jP} - \mu_{t-jP}|$; unrolling sums a geometric series.

At $P = 12, \alpha = 0.1$ the constant is $0.574 \tau + 0.394 \Delta_\mu$ plus a residual that contracts by $\theta = 0.28$ per block: for a tracked item with steady rate ($\Delta_\mu = 0$) and dispersion within τ at every block, letting J grow leaves $0.574 \tau \approx 1.28$ against $\mu_{\min} = 8$, a limiting statement whose residual after J blocks is θ^J times the error J blocks back. The estimator is thus accurate over the stable regime that defines the task; elsewhere the kernel mismatch is a transient at the ends of a run, measured as detection lag in the appendix of this document, and α is chosen with the drift term in Section V-B of the main paper.

E. Approximation Error of the MAD Estimator

CADENCE maintains $\widehat{\text{mad}}_t(u) = (1 - \alpha) \widehat{\text{mad}}_{t-1}(u) + \alpha |x_t(u) - \hat{\mu}_t(u)|$. Two quantities control its error: the centering drift $\Delta_t(u) \triangleq \max_{k < P} |\hat{\mu}_{t-k}(u) - \mu_t(u)|$, and the second-order dispersion $D_t(u) \triangleq \frac{1}{P} \sum_{k < P} |x_{t-k}(u) - \mu_t(u)| - \text{MAD}_t(u)$, the mean absolute deviation of the deviations, small for the regularly fluctuating items the gate admits.

Theorem 3 (MAD estimation error). *Under Assumption 1, for any item u tracked throughout epochs $t - P, \dots, t$, so that the value committed for each epoch of the window is the item's full per-epoch quantity,*

$$\begin{aligned} |\widehat{\text{mad}}_t(u) - \text{MAD}_t(u)| &\leq P \kappa_P(\alpha) D_t(u) + \Delta_t(u) \\ &\quad + \theta |\widehat{\text{mad}}_{t-P}(u) - \text{MAD}_t(u)|. \end{aligned}$$

Proof idea. Unroll as in (4) with summands $|x_{t-k} - \hat{\mu}_{t-k}|$ and compare each against $y_k = |x_{t-k} - \mu_t|$: the reverse triangle inequality bounds the difference by Δ_t , and writing $y_k = \text{MAD}_t + \eta_k$ with $\sum_k \eta_k = 0$, $\sum_k |\eta_k| = P D_t(u)$, the argument of Theorem 2 yields the remaining terms.

A lemma in the appendix of this document gives $D_t(u) \leq 2(P - 2)P^{-1} \text{MAD}_t(u)$ (attained), so the first term is at most $0.69 \text{MAD}_t(u)$, or 1.54 at dispersion τ . The drift term Δ_t compares the last P mean estimates against the *current* window mean. Theorem 2 does not bound Δ_t , and every argument we found that does yields a constant above τ at the operating point. The MAD bound therefore carries no constant here: it stays data-dependent through D_t and Δ_t , and we report it as measured.

Table II drives the estimator recursions with the exact per-epoch series of the 2,000 most frequent items per trace (no table and no collisions, so tracking is continuous by construction), at every item-epoch active throughout the window, 3.07M in all. Neither bound is violated; the mean bound lies below $\mu_{\min} = 8$ on 97–99% of item-epochs and the MAD bound below

τ on 52–82%. The mean bound averages 1.2–1.7 and the measured error 0.38–0.62 against $\mu_{\min}$, where the $B_{\max}$ bound is 93–3,064 and exceeds $\mu_{\min}$ everywhere. One caveat is the start: $\widehat{\text{mad}}_0 = 0$ leaves weight $(1 - \alpha)^N$ on 0 after N commits, so $\widehat{\text{mad}}$ is 28% low at $N = P = 12$, which alone already lets an item with true MAD 1.39τ clear the gate at qualification; the appendix of this document measures the cost of removing the bias. The drift term of Theorem 3 is charged in full, which is conservative: the kernel places weight $1 - \theta = 0.72$ on the P in-window deviations, so the same argument gives $(1 - \theta)\Delta_t(u)$. Re-running the same recursions with that term tightens the MAD bound by about a fifth, to 1.68, 2.09, 1.85 and 2.51, below τ on three of the four traces rather than one, and it is violated on none of the 3.07M item-epochs. The lemma’s constant is attained in the data, but the typical item is far from it: D_t/MAD_t averages 0.52–0.61 across the traces against the 1.67 it allows.

F. Collision Robustness and SSI Preservation

Call a bucket *protected* when its resident satisfies all three gates, $b.S \geq P$, $b.\mu \geq \mu_{\min}$ and $b.\text{mad} \leq \tau$. Recall $\text{score}(b) = b.S/P + b.\mu/\mu_{\min} - b.\text{mad}/\tau$, the collision score: a non-protected victim with $\text{score}(b) \leq 0$ is replaced deterministically, otherwise evicted with probability $p_{\text{evict}}(b) = 2^{-\lambda \text{score}(b)}$ ($\lambda > 0$).

Theorem 4 (Protected buckets are collision-invariant). *Suppose a bucket b satisfies $b.S \geq P$, $b.\mu \geq \mu_{\min}$, $b.\text{mad} \leq \tau$ at the time it is inspected. Then b is excluded from victim selection by Algorithm 1 and therefore cannot be evicted by a collision event unless, after a later rollover, at least one of the three inequalities ceases to hold.*

Proof idea. Lines 18–19 of Algorithm 1 skip such a bucket before it can be scored or assigned to *best*, and only *best* is reinitialized; the test is re-evaluated after synchronization on every scan.

Theorem 5 (Exponential suppression of eviction). *Let u be a tracked item that has not yet become protected. Suppose that during a time interval of interest, every time the bucket storing u is selected as the best victim candidate, its score is lower-bounded by $q_0 > 0$. If this happens at most C_u times, then $\Pr[u \text{ is evicted during the interval}] \leq C_u 2^{-\lambda q_0}$, and, conditioning on the history before each event, $\Pr[u \text{ survives all } C_u \text{ candidate events}] \geq (1 - 2^{-\lambda q_0})^{C_u}$.*

Proof idea. With score at least $q_0 > 0$ the deterministic branch is skipped, so each candidate event evicts u with probability $2^{-\lambda \text{bestScore}} \leq 2^{-\lambda q_0}$; a union bound over at most C_u events gives the first claim and the chain rule the second. The survival form is always at least as tight; the union-bound form assumes nothing about the trials and is non-vacuous when $\lambda q_0 > \log_2 C_u$.

Admission. Retention leaves open the dual failure: if every probed bucket is protected the update is discarded (Lines 24–25), so an item that would qualify later can be starved. An ε -override that instead evicts the most recently qualified of the m incumbents with probability ε yields an admission guarantee (the appendix of this document): an item issuing A_u updates under all-protected probes is admitted with probability at least $1 - (1 - \varepsilon)^{A_u}$, while a protected entry’s per-collision eviction probability rises from 0 to at most ε . No ε improved recall on average (the appendix of this document), so our rule keeps $\varepsilon = 0$.

APPENDIX

A. Additional Analytical Results

1) *Admission under Saturation:* Theorems 4 and 5 are retention guarantees, and leave open the starvation noted in Section II-F. The ε -override replaces the unconditional discard: when all m probed buckets are protected, the most recently qualified of them, the one whose current qualification run began latest and is hence the least established, is evicted with probability ε .

Theorem 6 (Admission under saturation). *Let u be an item that is active in every epoch of an interval of W epochs, and suppose that at every update u issues while it is not resident all m probed buckets are protected. Under the ε -override, u is admitted within W epochs with probability at least $1 - (1 - \varepsilon)^{A_u}$, where A_u is the number of updates u issues during the interval. Moreover, a protected entry’s eviction probability per collision is at most ε , so Theorem 4 is weakened by at most this controllable amount per all-protected collision event.*

The guarantee is driven by A_u , the number of updates u issues, not by the number of epochs it spans. An item at the significance threshold issues $P\mu_{\min} = 96$ updates over P epochs, for which $\varepsilon = 10^{-3}$ gives an admission probability of only 0.09; the bound becomes strong at the 10^3 – 10^4 updates a packet-level SSI issues over P epochs, where the same ε gives 0.63 and above 0.9999. The main paper keeps $\varepsilon = 0$ because no value improved recall on average (Table V).

2) Second-Order Dispersion:

Lemma 7 (Second-order dispersion is controlled by the MAD). *For every P -epoch window, $D_t(u) \leq \frac{2(P-2)}{P} \text{MAD}_t(u)$, and the bound is attained.*

Proof: Let $y_k = |x_{t-k} - \mu_t| \geq 0$, with mean $M = \text{MAD}_t(u)$, so that $D_t = \frac{1}{P} \sum_k |y_k - M|$. Since $\sum_k (y_k - M) = 0$, $\sum_k |y_k - M| = 2 \sum_{y_k > M} (y_k - M)$. The deviations $x_{t-k} - \mu_t$ sum to zero, so the positive ones and the negative ones each carry

total absolute mass $PM/2$, and no single y_k exceeds $PM/2$. If at least two indices have $y_k > M$, then $\sum_{y_k > M} (y_k - M) \leq \sum_k y_k - 2M = PM - 2M$; if exactly one does, the sum is at most $PM/2 - M$, which is no larger for $P \geq 2$. Hence $\sum_k |y_k - M| \leq 2(P-2)M$ and $D_t \leq 2(P-2)M/P$. Two deviations of $\pm PM/2$ and $P-2$ zeros attain it, and are realizable as a non-negative activity sequence when $\mu_t(u) \geq PM/2$. ■

Without the zero-sum constraint the extremal configuration puts all mass on one index and the constant is $2(P-1)/P$; the constraint removes one index's worth. At $P = 12$ the first term of Theorem 3 is therefore at most $2(P-2)\kappa_P(\alpha) \text{MAD}_t(u) = 0.69 \text{MAD}_t(u)$.

3) *Initialization, Admission, and Lag: Per-Trace Detail:* This section reports the initialization, admission and detection-lag measurements deferred from the main paper.

Initialization. Table III compares the default start ($\hat{\mu}_0 = \mu_{\min}$, $\widehat{\text{mad}}_0 = 0$), the neutral start ($\hat{\mu}_0 = 0$, no correction), and the debiased variant (both estimators at 0, divided by $1 - (1 - \alpha)^N$ with a per-bucket count N of committed epochs) at 786 buckets (40 KB for the default variant; the debiased variant adds two fixed 4 KB lookup tables for the correction factors and occupies 48 KB), ten seeds. The neutral start does not isolate the initialization effect, because a fresh bucket's retention score $S/P + \hat{\mu}/\mu_{\min} - \widehat{\text{mad}}/\tau$ then starts at 0 and it is evicted before it can qualify. The debiased variant raises precision on every trace and lowers recall, and F1 never improves by more than 0.001; the bias was acting as a de facto relaxation of the stability threshold.

TABLE III
INITIALIZATION VARIANTS AT 786 BUCKETS (40 KB DEFAULT, 48 KB DEBIASED): RECALL / PRECISION / F1.

Trace	default	neutral (F1)	debiased
CAIDA	0.895 / 0.921 / 0.908	0.001	0.853 / 0.934 / 0.892
Campus	0.672 / 0.784 / 0.724	0.043	0.651 / 0.812 / 0.723
Zipf	0.743 / 0.727 / 0.735	0.002	0.742 / 0.729 / 0.735
WebDocs	0.487 / 0.693 / 0.572	0.005	0.369 / 0.750 / 0.494

Saturation. Table IV gives, per trace and budget, the percentage of updates at which every probed bucket is protected (the discard branch), together with the peak number of SSIs concurrent in one epoch. The rate is highest at the smallest budgets; the ratio of concurrent SSIs to buckets exceeds one only on WebDocs below 10 KB.

TABLE IV
DISCARD-BRANCH RATE (% OF UPDATES) AND PEAK CONCURRENT SSIs.

Trace	peak	5	10	20	40	60	80	100 KB
CAIDA	18	0.678	0.215	0.053	0.014	0.006	0.004	0.004
Campus	17	0.323	0.150	0.037	0.007	0.002	0.002	0.001
Zipf	22	0.962	0.191	0.033	0.005	0.003	0.001	0.001
WebDocs	183	0.685	0.383	0.165	0.052	0.023	0.015	0.009

The ε -override. Table V sweeps ε at 5 and 10 KB. A qualified entry embodies P epochs of accumulated continuity, so evicting one discards that and forces re-qualification from scratch while unprotected; the values of ε large enough to make Theorem 6 meaningful therefore cost recall, and the small ones are within the seed spread.

TABLE V
RECALL UNDER THE ε -OVERRIDE.

Trace	KB	$\varepsilon = 0$	10^{-5}	10^{-4}	10^{-3}
CAIDA	5	0.790	0.785	0.763	0.708
CAIDA	10	0.838	0.835	0.837	0.818
Campus	5	0.470	0.469	0.478	0.448
Campus	10	0.540	0.540	0.541	0.547
WebDocs	5	0.210	0.213	0.201	0.185
WebDocs	10	0.319	0.317	0.315	0.305

Detection lag. Table VI gives the onset lag (epochs from the start of a ground-truth SSI run until the item is first reported) and the trailing lag (epochs the item is still reported after the run ends), as means over the detected runs with the median and 90th percentile, at 100 KB. Entry is gated by the exact streak, so the median onset is 0 on every trace; exit is gated by whichever descriptor fails first: a run that simply stops resets the streak at once, which is why the median trailing lag is also 0, whereas an item that keeps arriving leaves only once $\widehat{\text{mad}}$ crosses τ , several epochs later. That lateness is the tail in the means and p90s, and the transient counterpart of the kernel mismatch bounded in Theorem 2.

TABLE VI
DETECTION LAG IN EPOCHS AT 100 KB: MEAN (MEDIAN, P90).

Trace	onset	trailing
CAIDA	0.77 (0, 2)	6.38 (0, 13)
Campus	0.59 (0, 2)	1.72 (0, 4)
Zipf	0.34 (0, 1)	1.93 (0, 6)
WebDocs	0.84 (0, 3)	1.11 (0, 3)

4) Asymptotic Stability of the Mean Estimator:

Corollary 2 (Asymptotic EWMA deviation). *Under Assumption 1, suppose u is tracked throughout the N epochs since its insertion, $\alpha \in (0, 1)$, and there exist constants $\mu^* \in \mathbb{R}$ and $R > 0$ such that $|x_s(u) - \mu^*| \leq R$ at each of those epochs. Then CADENCE's EWMA mean estimate satisfies*

$$|\hat{\mu}_N(u) - \mu^*| \leq (1 - \alpha)^N |\hat{\mu}_0(u) - \mu^*| + R(1 - (1 - \alpha)^N),$$

and therefore $\limsup_{N \rightarrow \infty} |\hat{\mu}_N(u) - \mu^*| \leq R$, where N denotes the number of epochs since insertion.

Corollary 2 shows that, under bounded fluctuations of radius R around a reference demand level μ^* , the EWMA estimator is asymptotically stable: its deviation from μ^* remains bounded by R after the initialization transient vanishes. Note that this result is already free of $B_{\max}$ (it is stated in terms of the item's own fluctuation radius), and Theorems 2 and 3 bring the non-asymptotic bounds into the same form.

5) Collision Cascade:

Lemma 8 (Collision cascade under row occupancy). *Let $\rho_i \in [0, 1]$ denote the fraction of non-empty buckets in row i at a given time. If each row hash h_i maps keys uniformly over the d buckets, the h_i are mutually independent, and the probed key is not currently resident and hashes independently of the occupancy pattern, then, conditionally on that pattern,*

$$\Pr[\text{all } m \text{ candidate buckets are occupied}] = \prod_{i=1}^m \rho_i \leq \rho^m,$$

where $\rho \triangleq \max_{1 \leq i \leq m} \rho_i$.

6) *End-to-End Detection Guarantee:* We now combine the descriptor-estimation results with the collision analysis. The margin an item must clear has to be a quantity that can be evaluated, so we take it from Corollary 1 rather than from Theorem 2: the right-hand side of Theorem 2 contains $\theta |\hat{\mu}_{t-P} - \mu_t|$, which is a function of the sketch's internal state and is not bounded by anything in that theorem. Fix a horizon $J \geq 1$ and adopt Corollary 1's hypotheses on the exact statistics, $\text{MAD}_{t-jP}(u) \leq \tau$ for $0 \leq j < J$ and block drift at most Δ_μ ; residency over epochs $t - JP, \dots, t$ is not assumed but is the event whose probability Theorem 5 bounds, and the margins below are evaluated on that event. Define, with $\theta = (1 - \alpha)^P$,

$$\varepsilon_\mu \triangleq \frac{P \kappa_P(\alpha) \tau + \theta \Delta_\mu}{1 - \theta} + \theta^J R_J, \quad R_J \triangleq |\hat{\mu}_{t-JP}(u) - \mu_{t-JP}(u)| \quad (7)$$

$$\varepsilon_{\text{mad}}(t) \triangleq P \kappa_P(\alpha) D_t(u) + \Delta_t(u) + \theta |\widehat{\text{mad}}_{t-P}(u) - \text{MAD}_t(u)| \quad (8)$$

Corollary 1 bounds $|\hat{\mu}_t - \mu_t|$ by ε_μ , so (7) is a quantity a reader can evaluate from τ , Δ_μ and J up to the residual $\theta^J R_J$, which vanishes as J grows. At $P = 12, \alpha = 0.1$ and $\Delta_\mu = 0$, $\varepsilon_\mu \rightarrow 0.574\tau \approx 1.28$ as J grows, against the significance threshold $\mu_{\min} = 8$. With the $B_{\max}$ bound in place of Corollary 1 the corresponding quantity would be $B_{\max} \Xi_P(\alpha)$, at least 2.7×10^3 on CAIDA for any α and any $P \geq 8$; the margin condition of Corollary 3 would then be unsatisfiable in practice and the guarantee, while correct, empty.

We flag one asymmetry rather than leave it implicit: ε_{mad} in (8) still carries state-dependent terms, so the stability side of the margin condition is not reducible to constants the way the mean side is.

Corollary 3 (Margin-based SSI detection guarantee). *Consider a true SSI u at epoch t satisfying: (1) u is continuously active during the most recent P epochs, i.e., $x_{t-i}(u) > 0$ for all $i \in \{0, \dots, P-1\}$; (2) the exact statistics satisfy the hypotheses of Corollary 1 over the horizon J ($\text{MAD}_{t-jP}(u) \leq \tau$ for $0 \leq j < J$ and block drift at most Δ_μ) and the margin conditions*

$$\mu_t(u) \geq \mu_{\min} + \varepsilon_\mu, \quad \text{MAD}_t(u) \leq \tau - \varepsilon_{\text{mad}}(t);$$

with $\varepsilon_\mu, \varepsilon_{\text{mad}}$ as in (7)–(8), required to hold on every trajectory on which u is resident throughout epochs $t - JP, \dots, t$, since those margins are functions of the sketch's state; (3) u is tracked at epoch $t - JP$ and, during epochs $t - JP, \dots, t$, is selected

as the best victim candidate at most C_u times, each time with candidate score at least $q_0 > 0$. Then

$$\Pr[\text{CADENCE reports } u \text{ as an SSI at epoch } t] \geq 1 - C_u 2^{-\lambda q_0}.$$

Moreover, while the stored descriptors of u satisfy the SSI condition, collision-induced false negatives are excluded by Theorem 4, up to the ε -override of Theorem 6, which contributes at most ε per all-protected collision event.

7) *Durability Tracking*: The durability extension of the main paper reuses the streak machinery, so its guarantee follows from the results above without new assumptions. Let $g_t \triangleq [\hat{S}_t(u) \geq P \wedge \hat{\mu}_t(u) \geq \mu_{\min} \wedge \widehat{\text{mad}}_t(u) \leq \tau]$ denote the gate predicate on the stored descriptors at epoch t , and let the qualification streak Q be the number of consecutive most-recent epochs at which g_t holds.

Corollary 4 (Durability tracking up to the per-epoch margin). *Maintain the qualification streak Q as in the durability extension of the main paper. For a tracked item u under Assumption 1, Q equals exactly the number of consecutive most-recent epochs at which the stored descriptors $(\hat{S}_t, \hat{\mu}_t, \widehat{\text{mad}}_t)$ clear the three SSI gates. Consequently, if u is tracked from epoch $t_1 - JP$ onwards, at every epoch of a maximal interval $[t_1, t_2]$ the exact statistics meet the hypotheses and margin conditions of Corollary 3 (with its horizon J), and u is not evicted before becoming protected, then CADENCE reports a DSSI interval containing $[t_1, t_2]$, provided the run clears the durability threshold, $t_2 - t_1 + 1 \geq L$. More generally, for an item that satisfies Corollary 1's hypotheses at the epoch in question (so that ε_μ is available), the stored qualification g agrees with the true SSI qualification at every epoch whose exact $\mu_t(u), \text{MAD}_t(u)$ are bounded away from their thresholds by more than $\varepsilon_\mu, \varepsilon_{\text{mad}}(t)$; the reported maximal intervals therefore coincide with the true ones except at epochs within the margins, which may occur at interior as well as boundary epochs and can split, merge, or shift an interval.*

8) *Discussion*: One hypothesis in these statements is easy to misread. It is on the exact statistic, not on the gate: passing the gate certifies $\widehat{\text{mad}}_t \leq \tau$, which implies $\text{MAD}_t \leq \tau + \varepsilon_{\text{mad}}(t)$ rather than $\text{MAD}_t \leq \tau$, and using the gate here would make the argument circular; the bounds above are therefore stated on MAD_t and μ_t throughout. This is the sense in which the exponentially weighted descriptors are an adequate surrogate for the uniform P -epoch targets of Definition 1: not uniformly over all streams, but over the stable regime that defines the task, with the transient cost measured separately as detection lag in Appendix A, *Initialization, Admission, and Lag*.

B. Proofs

Proof of Theorem 1: Let ℓ_s denote the number of epochs committed while u is tracked, up to and including epoch s , so that $\ell_t \geq \ell$. We show by induction over the committed epochs that $\hat{S}_s(u) = \min\{\ell_s, S_s^*(u)\}$. Before any epoch of u has been committed, CADENCE initializes the streak to $0 = \min\{0, S^*\}$. When the bucket is next synchronized so as to commit epoch s (by a lazy rollover on the following Update, or by the periodic Query sweep if u is idle), the rollover commits the completed weight $x_s(u)$: if $x_s(u) > 0$, the rollover increments the stored streak to $\hat{S}_{s-1}(u) + 1 = \min\{\ell_{s-1} + 1, S_{s-1}^*(u) + 1\} = \min\{\ell_s, S_s^*(u)\}$, since an active epoch extends the exact streak by one; if $x_s(u) = 0$, it resets the streak to $0 = S_s^*(u)$. Since each completed epoch is committed exactly once under Assumption 1 (which forces $t - b.T = 1$ at every rollover, an idle epoch with $x_s(u) = 0$ being committed by the Query sweep), no epoch is skipped or double-counted. At epoch t the hypothesis $\ell \geq S_t^*(u)$ gives $\hat{S}_t(u) = S_t^*(u)$: the whole current streak, and the inactive epoch preceding it when $\ell > S_t^*(u)$, was committed while u was tracked.

If instead the bucket is reinitialized by a collision at some epoch t' in the interval, the streak is reset to 0 and the conclusion fails for $t \geq t'$; the hypothesis is thus not merely technical. Theorem 4 shows that this cannot happen once u has qualified, and Theorem 5 bounds the probability that it happens before then. ■

Proof of Theorem 2: Under Assumption 1 each completed epoch is committed exactly once, so unrolling the estimator recursion over the P epochs $t - P + 1, \dots, t$ is valid and gives

$$\hat{\mu}_t(u) = (1 - \alpha)^P \hat{\mu}_{t-P}(u) + \alpha \sum_{k=0}^{P-1} (1 - \alpha)^k x_{t-k}(u) \quad (9)$$

Put

$$c_k \triangleq \alpha(1 - \alpha)^k - \frac{1}{P}, \quad 0 \leq k \leq P - 1$$

and note the identity

$$\sum_{k=0}^{P-1} c_k = \alpha \frac{1 - (1 - \alpha)^P}{1 - (1 - \alpha)} - 1 = (1 - (1 - \alpha)^P) - 1 = -(1 - \alpha)^P \quad (10)$$

Subtracting $\mu_t(u) = \frac{1}{P} \sum_{k=0}^{P-1} x_{t-k}(u)$ from (9),

$$\hat{\mu}_t(u) - \mu_t(u) = (1 - \alpha)^P \hat{\mu}_{t-P}(u) + \sum_{k=0}^{P-1} c_k x_{t-k}(u)$$

Now write $x_{t-k}(u) = \mu_t(u) + \delta_k$, where by construction

$$\sum_{k=0}^{P-1} \delta_k = 0, \quad \sum_{k=0}^{P-1} |\delta_k| = P \text{MAD}_t(u) \quad (11)$$

the second equality being the definition of the P -epoch mean absolute deviation. Using (10),

$$\sum_{k=0}^{P-1} c_k x_{t-k}(u) = \mu_t(u) \sum_{k=0}^{P-1} c_k + \sum_{k=0}^{P-1} c_k \delta_k = -(1 - \alpha)^P \mu_t(u) + \sum_{k=0}^{P-1} c_k \delta_k$$

and therefore the *exact identity*

$$\hat{\mu}_t(u) - \mu_t(u) = (1 - \alpha)^P (\hat{\mu}_{t-P}(u) - \mu_t(u)) + \sum_{k=0}^{P-1} c_k \delta_k \quad (12)$$

No inequality has been used so far. It remains to bound $\sum_k c_k \delta_k$ over the set (11), namely $\{\sum_k \delta_k = 0, \sum_k |\delta_k| = Z\}$ with $Z = P \text{MAD}_t(u)$. Both constraints matter, and using only the second (as Hölder with the ℓ^∞ - ℓ^1 pairing does) is loose by a factor of 1.51 at the paper's settings. Split $\delta = \delta^+ - \delta^-$ into positive and negative parts; the two constraints give $\sum_k \delta_k^+ = \sum_k \delta_k^- = Z/2$, hence

$$\sum_{k=0}^{P-1} c_k \delta_k = \sum_k c_k \delta_k^+ - \sum_k c_k \delta_k^- \leq \frac{Z}{2} \left(\max_k c_k - \min_k c_k \right) = P \kappa_P(\alpha) \text{MAD}_t(u)$$

with $\kappa_P(\alpha)$ as in (5). The same argument applied to $-\delta$ bounds the absolute value. The bound is attained by placing mass $Z/2$ at $\arg \max_k c_k$ and $-Z/2$ at $\arg \min_k c_k$, so no smaller constant is available over this constraint set (the extremal series is realizable as a non-negative activity sequence when $\mu_t(u) \geq P \text{MAD}_t(u)/2$); substituting into (12) and applying the triangle inequality proves the theorem.

Remark (the $B_{\max}$ bound, and why it is vacuous). With

$$w_k = \alpha(1 - \alpha)^k, \quad v_k = \frac{1}{P} \ (k < P), \ 0 \ (k \geq P), \quad \Xi_P(\alpha) \triangleq \sum_{k \geq 0} |w_k - v_k| = \sum_{k=0}^{P-1} \left| \alpha(1 - \alpha)^k - \frac{1}{P} \right| + (1 - \alpha)^P,$$

the $B_{\max}$ argument bounds $|\sum_k (w_k - v_k) x_{t-k}|$ by $B_{\max} \Xi_P(\alpha)$: it paired ℓ^1 on the kernel with ℓ^∞ on the *levels* x_{t-k} . The argument above pairs the kernel against the *deviations* δ_k and uses $\sum_k \delta_k = 0$. It is the change of the quantity being bounded, $\text{MAD}_t(u)$ in place of $B_{\max}$, that removes the dependence on the stream's absolute scale.

The constant itself did not improve much, and we state this plainly because it is easy to misread the result as a tighter constant. Numerically $\Xi_{12}(0.1) = 0.6115$ against $P \kappa_{12}(0.1) = 0.4117$; the two are within a factor of 1.5. Retuning would not rescue that bound either: at $P = 12$ the minimum of $\Xi_P(\alpha)$ over α is 0.5635 (attained at $\alpha = 0.1283$; its value at $\alpha = 0.13$ is 0.5636), and over the horizons SSIL targets ($P \geq 8$) the minimum is 0.558. On CAIDA, where $B_{\max} = 5010$, the $B_{\max}$ bound is therefore at least 2.7×10^3 for any α and any $P \geq 8$, and 3.06×10^3 at our defaults, against a significance threshold of 8; the first term of Theorem 2 is $P \kappa_P(\alpha) \text{MAD}_t(u) \leq 0.93$ for an item whose true dispersion clears τ .

Note also that $(1 - \alpha)^P = 0.2824$, or 46% of $\Xi_P(\alpha)$, is the weight the exponential kernel places outside the P -epoch window. Theorem 2 does not remove that weight; it carries it as the data-dependent term $\theta |\hat{\mu}_{t-P} - \mu_t|$, which is why a numerical guarantee requires Corollary 1 and its hypotheses rather than Theorem 2 alone. ■

Proof of Corollary 1: Write $\theta = (1 - \alpha)^P$ and, for $j \geq 0$, $e_j \triangleq |\hat{\mu}_{t-jP}(u) - \mu_{t-jP}(u)|$. Applying (12) at epoch $t - jP$ and inserting μ_{t-jP} into the trailing term,

$$|\hat{\mu}_{t-(j+1)P} - \mu_{t-jP}| \leq e_{j+1} + |\mu_{t-(j+1)P} - \mu_{t-jP}| \leq e_{j+1} + \Delta_\mu$$

so that, using $\text{MAD}_{t-jP}(u) \leq \tau$,

$$e_j \leq P \kappa_P(\alpha) \tau + \theta (e_{j+1} + \Delta_\mu), \quad 0 \leq j < J$$

Unrolling this recursion J times and summing the geometric series $\sum_{j \geq 0} \theta^j = (1 - \theta)^{-1}$,

$$e_0 \leq (P \kappa_P(\alpha) \tau + \theta \Delta_\mu) \sum_{j=0}^{J-1} \theta^j + \theta^J e_J \leq \frac{P \kappa_P(\alpha) \tau + \theta \Delta_\mu}{1 - \theta} + \theta^J e_J$$

which is the claim. At $P = 12, \alpha = 0.1$ one has $\theta = 0.2824$ and $P \kappa_P(\alpha) = 0.4117$, so the leading coefficient is $0.4117136/0.7175705 = 0.5737606$ and the residual contracts by a factor 0.2824 per block. ■

Proof of Corollary 2: Writing the recursion with N epochs since insertion (the same expansion as above with lag k replaced by $N - i$),

$$\hat{\mu}_N(u) = (1 - \alpha)^N \hat{\mu}_0(u) + \alpha \sum_{i=1}^N (1 - \alpha)^{N-i} x_i(u).$$

Subtracting μ^* and using $1 = (1 - \alpha)^N + \alpha \sum_{i=1}^N (1 - \alpha)^{N-i}$,

$$\begin{aligned} \hat{\mu}_N(u) - \mu^* &= (1 - \alpha)^N (\hat{\mu}_0(u) - \mu^*) \\ &\quad + \alpha \sum_{i=1}^N (1 - \alpha)^{N-i} (x_i(u) - \mu^*). \end{aligned}$$

Applying the triangle inequality and $|x_i(u) - \mu^*| \leq R$,

$$|\hat{\mu}_N(u) - \mu^*| \leq (1 - \alpha)^N |\hat{\mu}_0(u) - \mu^*| + \alpha R \sum_{i=1}^N (1 - \alpha)^{N-i}.$$

Since $\alpha \sum_{i=1}^N (1 - \alpha)^{N-i} = 1 - (1 - \alpha)^N$, the stated bound follows; taking lim sup yields the second claim. ■

Proof of Theorem 3: Set

$$a_k = |x_{t-k}(u) - \hat{\mu}_{t-k}(u)|, \quad b_k = |x_{t-k}(u) - \mu_t(u)|, \quad 0 \leq k \leq P-1$$

so that $\frac{1}{P} \sum_k b_k = \text{MAD}_t(u)$. Unrolling the MAD recursion over the same P epochs, exactly as in (9),

$$\widehat{\text{mad}}_t(u) = (1 - \alpha)^P \widehat{\text{mad}}_{t-P}(u) + \alpha \sum_{k=0}^{P-1} (1 - \alpha)^k a_k$$

Subtracting $\text{MAD}_t(u)$ and adding and subtracting $\sum_k \alpha (1 - \alpha)^k b_k$,

$$\begin{aligned} \widehat{\text{mad}}_t(u) - \text{MAD}_t(u) &= (1 - \alpha)^P \widehat{\text{mad}}_{t-P}(u) \\ &\quad + \underbrace{\sum_{k=0}^{P-1} \alpha (1 - \alpha)^k (a_k - b_k)}_{(\mathbf{A})} + \underbrace{\sum_{k=0}^{P-1} c_k b_k}_{(\mathbf{B})} \end{aligned}$$

with c_k as in the proof of Theorem 2.

For (A), the reverse triangle inequality gives $|a_k - b_k| \leq |\hat{\mu}_{t-k}(u) - \mu_t(u)| \leq \Delta_t(u)$ for every in-window lag $0 \leq k \leq P-1$, and $\alpha \sum_{k=0}^{P-1} (1 - \alpha)^k = 1 - (1 - \alpha)^P \leq 1$, so $|(\mathbf{A})| \leq (1 - (1 - \alpha)^P) \Delta_t(u) \leq \Delta_t(u)$.

For (B), write $b_k = \text{MAD}_t(u) + \eta_k$. By construction $\sum_k \eta_k = 0$ and, by the definition of the second-order dispersion D_t , $\sum_k |\eta_k| = P D_t(u)$. Using (10) and then the zero-sum split argument of Theorem 2 (the deviations η_k sum to zero, so their positive and negative parts each carry mass $P D_t(u)/2$),

$$(\mathbf{B}) = -(1 - \alpha)^P \text{MAD}_t(u) + \sum_{k=0}^{P-1} c_k \eta_k, \quad \left| \sum_k c_k \eta_k \right| \leq P \kappa_P(\alpha) D_t(u)$$

The term $-(1 - \alpha)^P \text{MAD}_t(u)$ combines with $(1 - \alpha)^P \widehat{\text{mad}}_{t-P}(u)$ to give $(1 - \alpha)^P (\widehat{\text{mad}}_{t-P}(u) - \text{MAD}_t(u))$. Collecting the three contributions and applying the triangle inequality yields

$$|\widehat{\text{mad}}_t(u) - \text{MAD}_t(u)| \leq P \kappa_P(\alpha) D_t(u) + \Delta_t(u) + (1 - \alpha)^P |\widehat{\text{mad}}_{t-P}(u) - \text{MAD}_t(u)|$$

as claimed. As with Theorem 2, $B_{\max}$ does not appear: the estimator is accurate whenever the centering does not drift (which Theorem 2 and Corollary 1 control) and the fluctuations are themselves regular, i.e. $D_t(u)$ is small.

Remark (initialization). If $\widehat{\text{mad}}_0(u) = 0$ and no debiasing is applied, then after N committed epochs a weight $(1 - \alpha)^N$ rests on the value 0, so $\widehat{\text{mad}}$ is biased low by the factor $1 - (1 - \alpha)^N$, which is where the gate-clearing MAD $\tau / (1 - (1 - \alpha)^P) = 1.39 \tau$ comes from. Starting both estimators at 0 and dividing by $1 - (1 - \alpha)^N$ removes the weight deficit exactly; for the MAD a residual centering bias remains, because the deviations are taken against $\hat{\mu}_i$, which is itself in transient. CADENCE does *not* do this by default (it keeps $\hat{\mu}_0 = \mu_{\min}$, $\widehat{\text{mad}}_0 = 0$, and the debiased variant is reported as an ablation), and the correction would in any case not touch the trailing term of Theorem 2: $\theta|\hat{\mu}_{t-P} - \mu_t|$ is the weight the exponential kernel places outside the window acting on the data, not an artefact of $\hat{\mu}_0$ once the item has been tracked for several windows; at $N = P$ exactly, $\hat{\mu}_{t-P}$ is still $\hat{\mu}_0$ and the term is $\theta|\mu_{\min} - \mu_t|$. The default start and the division are also incompatible: the division is correct only for a zero start. ■

Proof of Lemma 8: CADENCE probes exactly one candidate bucket per row. For row i , the probability that the hashed bucket is occupied equals the row occupancy fraction ρ_i . Under independence across rows, the probability that all m candidate buckets are occupied is the product $\prod_{i=1}^m \rho_i$. Since each $\rho_i \leq \rho$, we further obtain $\prod_{i=1}^m \rho_i \leq \rho^m$. This proves the claim. ■

Proof of Theorem 4: The claim follows directly from the protection gate of the `Update` algorithm in the main paper. Once a resident satisfies the three SSI inequalities, the protection rule issues a `continue`, which bypasses the candidate-scoring step; the bucket is therefore never assigned to `best`, and the subsequent replacement block reinitializes only `best`. Hence the algorithm never considers that bucket as a replacement candidate in the current collision scan. The same logic applies to every future scan as long as the SSI inequalities remain true after synchronization.

With the ε -override of Theorem 6 enabled, the conclusion weakens by exactly a controllable amount: a protected bucket is exposed only on collision events at which every probed bucket is protected, and then only if it is the most recently qualified of them (the one whose current qualification run began latest, hence the least established), so its eviction probability is at most ε per such event and at most $C\varepsilon$ over C of them by the union bound. Setting $\varepsilon = 0$, our rule, recovers the statement of probability zero. ■

Proof of Theorem 5: At each candidate event the bucket's score equals $\text{bestScore} \geq q_0 > 0$, so the deterministic branch ($\text{bestScore} \leq 0$) never applies and u can be evicted only through the probabilistic trial, with probability $2^{-\lambda \text{bestScore}} \leq 2^{-\lambda q_0}$ by monotonicity of $x \mapsto 2^{-\lambda x}$. Writing $\mathcal{F}_{\ell-1}$ for the history before u 's ℓ -th candidate event, that event evicts u with probability at most $2^{-\lambda q_0}$ conditional on $\mathcal{F}_{\ell-1}$; summing over the at most C_u events gives

$$\Pr[\text{at least one eviction}] \leq \sum_{\ell=1}^{C_u} 2^{-\lambda q_0} = C_u 2^{-\lambda q_0}$$

Writing p_ℓ for the eviction probability at event ℓ conditioned on the history before it, iterated conditioning gives $\Pr[\text{survive all}] = \mathbb{E}[\prod_{\ell=1}^{C_u} (1 - p_\ell)] \geq (1 - 2^{-\lambda q_0})^{C_u}$, because each $p_\ell \leq 2^{-\lambda q_0}$. ■

Proof of Theorem 6: Consider the collision events at which u probes its m candidate buckets and finds all of them protected. Our rule discards the update at such an event, so if the condition persists u is never inserted, however long it remains active: the discard is deterministic, and no sequence of updates can overcome it. This is the starvation mode the theorem removes.

Under the ε -override, each such event instead evicts the most recently qualified of the m protected incumbents with probability ε , using a fresh draw from the update stream's random source. The draws at distinct events are independent by construction. Let A_u be the number of updates u issues during the interval, as in the theorem. On the event that u is never admitted it is never resident, so each of those updates contributes one such event under the theorem's hypothesis, and admission ends the sequence. Writing Z_ℓ for the indicator that the ℓ -th draw succeeds,

$$\Pr[u \text{ not admitted}] = \Pr\left[\bigcap_{\ell=1}^{A_u} \{Z_\ell = 0\}\right] = (1 - \varepsilon)^{A_u}$$

so u is admitted with probability at least $1 - (1 - \varepsilon)^{A_u}$. Because u is active in every epoch of the interval it issues at least one update per epoch, whence $A_u \geq W$ and the weaker but parameter-free bound $1 - (1 - \varepsilon)^W$ also holds.

For the second claim, fix a protected entry b . It is exposed only at all-protected events among its own probes, and at such an event it is evicted only if it is the most recently qualified incumbent, so its per-event eviction probability is at most ε ; this is the statement already recorded in the proof of Theorem 4. Hence the admission guarantee and the retention guarantee trade off through the single parameter ε , and not symmetrically: the admission side grows with A_u , while the cost side, at most ε per collision for a qualified entry, does not vary with the item at all. ■

Proof of Corollary 3: Condition on the event that u is not evicted during epochs $t - JP, \dots, t$; u is exposed only at its at most C_u candidate selections, each with score at least q_0 , so by Theorem 5 this event has probability at least $1 - C_u 2^{-\lambda q_0}$. On this event u is tracked continuously over epochs $t - JP, \dots, t$, so by Assumption 1 every completed epoch is committed exactly once, and Corollary 1 applies with $R_J = |\hat{\mu}_{t-JP}(u) - \mu_{t-JP}(u)|$. By the induction of Theorem 1 the stored streak equals the number of consecutive active epochs committed while u is tracked; since u is active over the most recent P epochs and at least

$JP \geq P$ epochs have been committed while tracked, $\hat{S}_t(u) = \min\{\ell_t, S_t^*(u)\} \geq P$. By Theorem 2 (with Corollary 1 in the stable regime) and Theorem 3, together with margin conditions exceeding the error terms (7)–(8), the stored mean and MAD satisfy $\hat{\mu}_t(u) \geq \mu_{\min}$ and $\widehat{\text{mad}}_t(u) \leq \tau$. Hence u passes all three gates and is reported by `Query`. Taking the probability of the conditioning event proves the bound. The margin is taken from Corollary 1, so it inherits that corollary’s hypotheses: residency over J blocks, $\text{MAD} \leq \tau$ at each block, and block drift at most Δ_μ . ■

Proof of Corollary 4: The gate predicate g is a deterministic function of the stored descriptors, which Assumption 1 refreshes exactly once per committed epoch. The recurrence maintaining Q is identical in form to the activity-streak recurrence of Theorem 1 with the event “ $x_t(u) > 0$ ” replaced by “ g holds,” so the same induction shows Q counts exactly the consecutive most-recent epochs at which g is true. The two-sided error bounds of Theorems 2 and 3, together with the streak invariant of Theorem 1, ensure that at every epoch whose exact $\mu_t(u), \text{MAD}_t(u)$ clear or fail their thresholds by more than $\varepsilon_\mu, \varepsilon_{\text{mad}}(t)$, the stored gate g agrees with the true SSI condition (the mean margin ε_μ is the Corollary 1 bound, so this presupposes that corollary’s hypotheses at the epoch). If the margins hold at every epoch of $[t_1, t_2]$ and u is not evicted before it qualifies, g is true at every epoch of $[t_1, t_2]$ and Theorem 4 keeps u resident from then on, so the stored run through t_1 covers $[t_1, t_2]$. The runs can otherwise differ only at in-margin epochs, which need not be run boundaries, and an interior in-margin epoch may split or merge runs. The protection and eviction accounting carry over unchanged from Theorems 4 and 5, since any bucket with $Q \geq 1$ satisfies the gates and is collision-invariant up to the ε -override. ■

C. Running Example of the Update Operation

We use a toy sketch ($m = 2$ rows, $d = 2$ buckets per row) with $P = 3$, $\mu_{\min} = 6$, $\tau = 2$, $\lambda = 4$, and $\alpha = \frac{1}{4}$ (small values for a legible walk-through; the defaults appear in Sections V-A and V-B of the main paper); each bucket stores $(K, T, C, S, \mu, \text{mad})$ and each scenario is independent. The two simple cases are immediate: a *match* (key resident in the current epoch) increments C , and an *empty insertion* writes $(K, T=t, C=1, S=0, \mu=\mu_{\min}, \text{mad}=0)$. We detail the two collision cases, giving the state before rollover as well as after.

Probabilistic replacement. A request from u_9 in epoch $t = 6$ probes $B(1, 1) = (K(u_1), T=5, C=8, S=2, \mu=7, \text{mad}=1)$ and $B(2, 1) = (K(u_2), T=5, C=3, S=1, \mu=4, \text{mad}=2)$. After lazy rollover $B(1, 1)$ becomes $S=3, \mu \approx 7.25, \text{mad} \approx 0.94$, so it is protected, while $B(2, 1)$ becomes $S=2, \mu \approx 3.75, \text{mad} \approx 1.69$, giving score $= S/P + \mu/\mu_{\min} - \text{mad}/\tau \approx 0.45$. Hence $\text{best} = B(2, 1)$ with $\text{bestScore} \approx 0.45 > 0$, so eviction is probabilistic with $p_{\text{evict}} = 2^{-\lambda \cdot \text{bestScore}} \approx 0.29$; on success $B(2, 1)$ is reinitialized for u_9 , otherwise the request is discarded.

Deterministic replacement. A request from u_{10} in epoch $t = 7$ probes $B(1, 2) = (K(u_4), T=6, C=0, S=0, \mu=1, \text{mad}=3)$ and $B(2, 2) = (K(u_6), T=6, C=4, S=1, \mu=4, \text{mad}=1)$. After lazy rollover their scores are ≈ -1.09 and ≈ 0.96 , so $\text{best} = B(1, 2)$ with $\text{bestScore} \approx -1.09 \leq 0$: the resident is evicted deterministically and the bucket is reinitialized for u_{10} .

Sweep commit. A bucket that receives no update during an epoch (bucket B in Figure 2 of the main paper) is not visited by any `Update`, so its elapsed epoch is committed by the periodic `Query` sweep (Algorithm 2): the sweep performs the same rollover, committing the idle epoch’s $x = 0$, which resets the streak to 0 and folds the zero into the estimators, exactly as an `Update`-triggered rollover would.

D. Why a Large α Under-reads the MAD

Section V-B of the main paper reports that F1 is an inverted-U in α and attributes the upper arm to the MAD estimate under-reading. The cause is the order in which `Update` commits the two estimators. The mean is committed first and the deviation is then taken against the *updated* mean, so with $\hat{\mu}_t = (1 - \alpha)\hat{\mu}_{t-1} + \alpha x_t$,

$$x_t - \hat{\mu}_t = (1 - \alpha)(x_t - \hat{\mu}_{t-1}), \quad (13)$$

and every deviation entering $\widehat{\text{mad}}$ is already damped by $(1 - \alpha)$. At $\alpha = 0.1$ the damping is 10% and the estimator tracks the window MAD closely; at $\alpha = 0.5$ it halves every deviation, which is why erratic items clear the stability gate and precision collapses while recall is unaffected. Theorem 3 does not miss this. Its drift term Δ_t is what charges the damping: rearranging the mean recurrence gives $\hat{\mu}_{t-k} - \mu_t = \alpha(x_{t-k} - \mu_t) + (1 - \alpha)(\hat{\mu}_{t-k-1} - \mu_t)$, so the per-epoch under-read is bounded by Δ_t and grows with α , which is why the bound loosens. What the theorem does not do is isolate the mechanism, which is what this section supplies.